29/04/26

Exercise 2 ; SQL Aggregate Functions &

Question 1

```
SELECT Dinstinct department
FROM students ;
```

Departments
IT
HR
~~IT~~
Finance

Question 2

```
SELECT department,
       age
FROM students ;
```

department	age
IT	20
IT	21
HR	22
Finance	23
	22

```
SELECT department,
       AVG (age) AS avg-age
FROM student
GROUP BY department;
```

department	avg-age
IT	20
HR	22
Finance	23

Question 3

```
SELECT department,
       CNT(~~students~~ *) AS cnt-student
FROM student
GROUP BY department H
HAVING COUNT (*) > 1;
```

department	~~COUNT~~ student-count
IT	2
HR	2

Question 4

```
SELECT    student_id,
          name,
          age,
          department
FROM      student
WHERE     age is BETWEEN 21 AND 23;
```

student_id	name	age	department
2	Bob	22	HR
5	EVE	22	HR

Question 5

```
SELECT (student_id),
        name,
        age,
        department

FROM    students
WHERE   department = 'IT' OR 'HR' > 21
```

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
5	EVE	22	HR

Question 6

```
SELECT      department,
            SUM (credits) AS total_credits
FROM courses
GROUP BY    department
HAVING SUM (credits) > 5;
```

department	total_credits
IT	11

Question 7

```
SELECT (*)
FROM    courses
WHERE   courses NOT HAVE 4 credits <>;
```

courses

Course_id	Course_name	department	credits
101	SQL Basics	IT	3
104	Excel	~~F~~ Finance	2
105	Statistics	HR	3

Question 8

```
SELECT      course_id,
            Course_name,
            Credit
FROM courses
ORDER BY    credits DESC
LIMIT       3
```

course_id	course_name	credits
103	I T	4
102	I T	4
101	I T	3

Question 9

```
SELECT    MAX(grade) AS max_grade,
~~FROM~~  ~~enrollment~~ MIN(grade) AS min_grade,
          AVG(grade) AS AVG_grade
FROM      enrollments
```

max_grade	min_grad	avg_grade
90	78	84,6

Question 10

```
SELECT    course_id,
          COUNT(enrollments) AS enrollment_count
FROM      enrollments
GROUP BY  course_id;
```

course_id	enrollment_count
101	1
102	2
103	3
104	4
105	5

Question 11

```
SELECT department,
       SUM(salary) AS total_salary,
       SUM(bonus) AS total_bonus
FROM salaries
GROUP BY department;
```

department	total_salary	total_bonus
HR	122 000	10 900
IT	109 000	7 900
Finance	70 000	6000

Question 12

```
SELECT department,
       AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55 000
```

department	avg-salary
IT	61 000

Question 13

```
SELECT    employee_id,
          name,
          salary,
          bonus,
          SUM(salary+bonus) AS total_compensation
FROM salaries
GROUP BY  employee_id,
          name,
          salary,
          bonus
HAVING    total_compensation > 60 000;
```

employee_id	name	salary	bonus	total_compensation
1	Tome	~~IT~~ 60 000	5000	65 000
3	Spike	70 000	6000	76 000
4	Tyke	62 000	5500	67 500

Question 14

```
SELECT    department,
          SUM(budget) AS total_budget,
          AVG(budget) AS avg_budget
FROM projects
GROUP BY  department
HAVING    avg_budget > 70 000
```

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

Question 15

```
SELECT (*)
FROM projects
WHERE budget BETWEEN 50 000 and 120 000
```

project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll system	Finance	80 000
3	Dash board	IT	150 000
5	HR Portal	HR	50 000